Experimental and Cohort Studies

APHS 20208 · Epidemiology & Biostatistics

Review: Association vs Causality

What is an **association**?

- The distribution of an outcome differs across levels of an exposure.
  - For example, the outcome may be more common in one group than another.
- Knowing something about **X** helps us predict something about **Y**.

Review: Association vs Causality

What distinguishes an **association** from a **causal relationship**?

- **Counterfactual definition:** The outcome would differ for the same person if their exposure had been different.
- **Intervention definition:** Intervening to change the exposure would change the outcome or its probability. Said another way, X is a cause of Y if we can change Y by manipulating X.
- **Helpful question**: If I changed X and held everything else constant, would I expect Y to change?

Review: Deterministic vs Probabilistic Causality

What distinguishes **deterministic** causality from **probabilistic** causality?

- **Deterministic:** Exposure **always** produces the outcome.
- **Probabilistic:** Exposure **increases the likelihood** of the outcome.

Two Categories of Analytic Studies

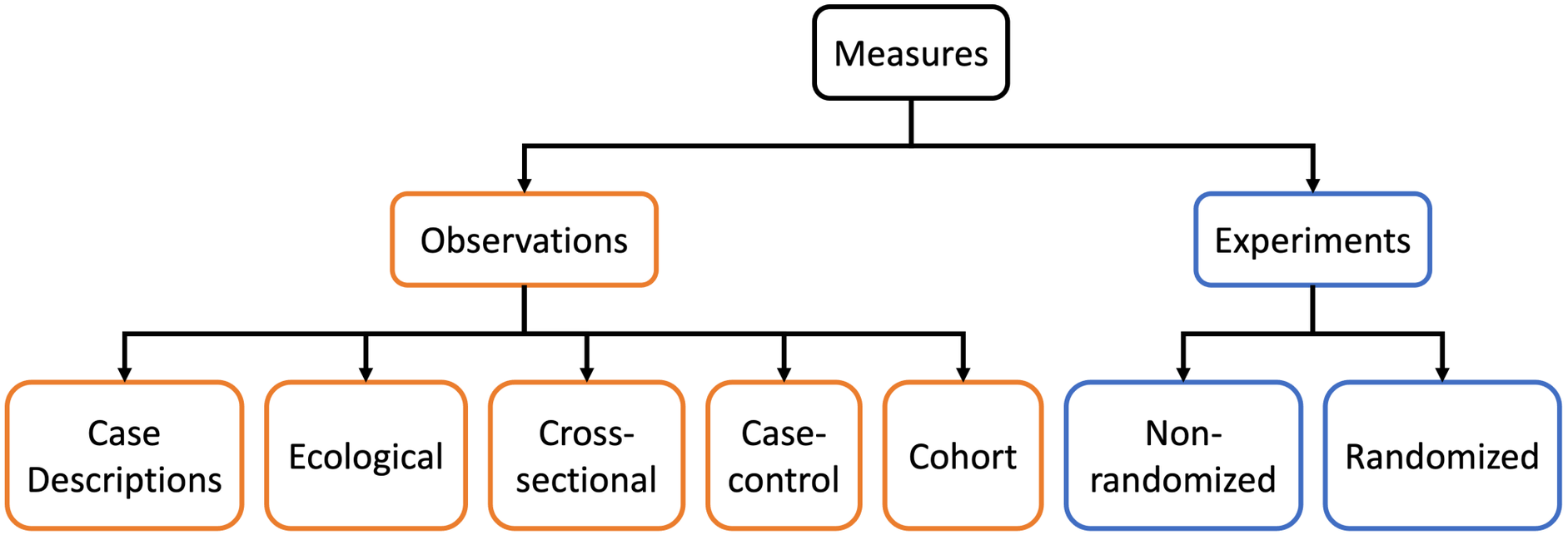
Observational design

- Investigator **does not assign exposure**
- Observes existing differences in exposure and outcome
- Examples include cross-sectional, cohort, and case-control studies

Experimental design

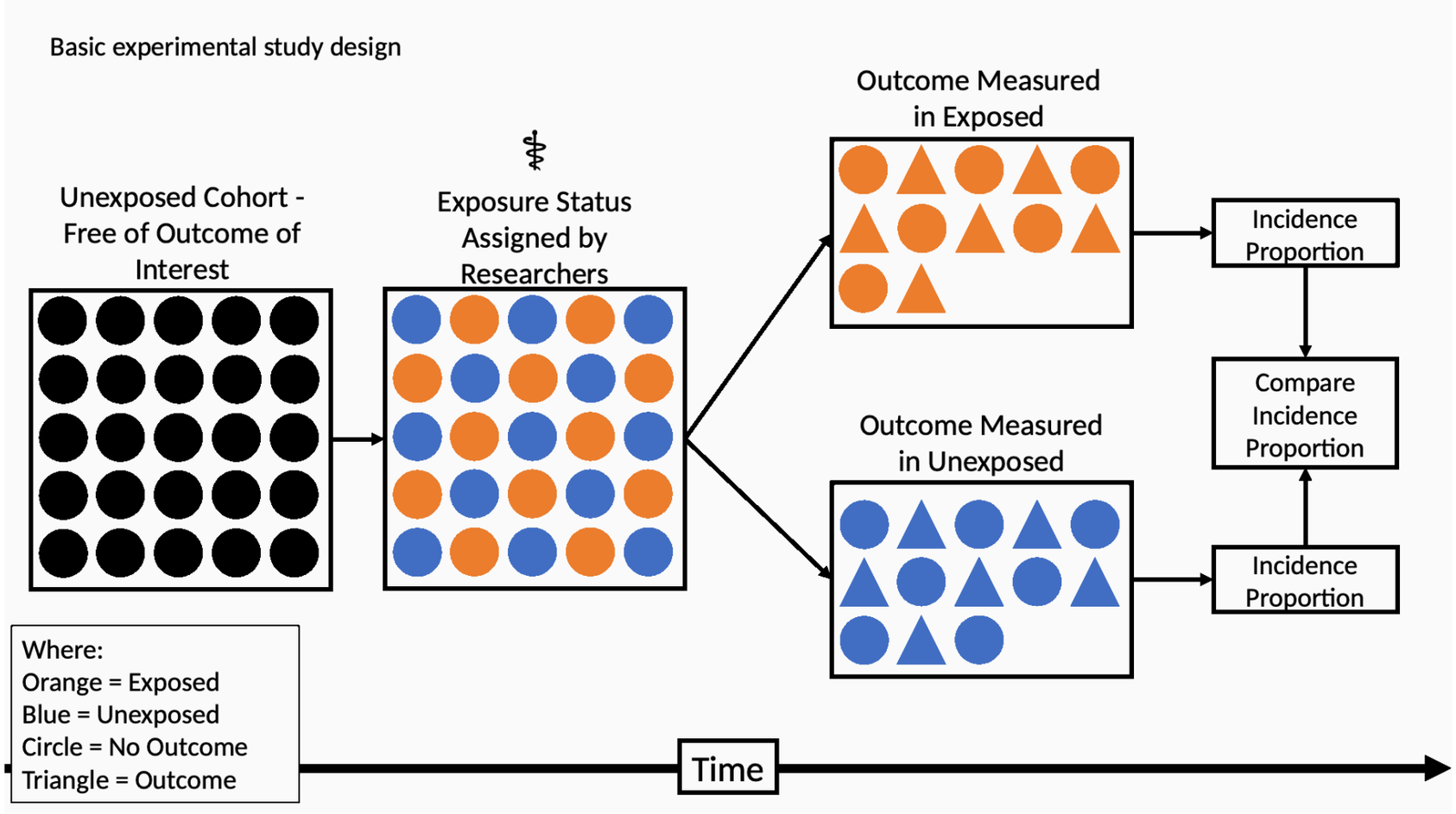
- Investigator **assigns exposure or intervention**
- Often uses **randomization**
- Examples include randomized controlled trials (RCTs)

Hierarchy of Study Designs



Typically less convincing → Typically more convincing

Basic Experimental Study Design



RCT: Randomization

Randomization assigns participants to treatment groups using a chance process.

Why randomization matters

- Breaks the link between **baseline causes of the outcome** and **treatment assignment**.
- Protects against confounding by **measured and unmeasured factors**, without needing to identify them all.
- Produces **exchangeable groups on average**: groups expected to have the same outcome risk if given the same treatment.

Take-home message

Randomization protects against **confounding on average**. Chance differences between groups can still occur in a particular trial.

Practice: Baseline Group Comparability

Participants were randomly assigned to aspirin (**8,517**) or placebo (**8,514**).

Baseline characteristic	Aspirin, n	Placebo, n	Ratio*
Past cigarette smoking	3,321	3,320	1.00
Current cigarette smoking	847	841	1.01
Diabetes mellitus	196	195	1.01

Ratio of the proportion with the characteristic in the aspirin group to the proportion in the placebo group; labeled RR in the source table.

1. What does a ratio close to **1** tell us about a baseline characteristic?
2. Do these rows describe treatment outcomes or participants **before treatment**?
3. Do similar proportions for these three characteristics establish that the groups are identical in every way?

Interpreting the Baseline Table

1. Ratios close to **1** indicate similar proportions with these **measured baseline characteristics**.
2. These rows contain baseline comparisons, so they describe participants **before treatment**.
3. Similarity in these measured characteristics does **not** establish similarity in every measured or unmeasured characteristic.
    - Randomization supports comparability **on average**; chance differences can remain in a particular trial.

Internal and External Validity

- **Internal validity:** Does the study provide an unbiased estimate of the association *among the people studied*?
  - Comparable groups at baseline, like the aspirin and placebo groups we just examined, support internal validity.
- **External validity:** Do the findings apply to people *outside the study*?
  - This depends on who enrolled. Would the aspirin results hold for people who differ from the trial participants in age, sex, or health?

RCTs: Strengths & Limitations

Strengths

- Randomization improves **group comparability**
- Strong **internal validity**
- Typically provides strongest evidence for the **causal effects** of interventions

Limitations

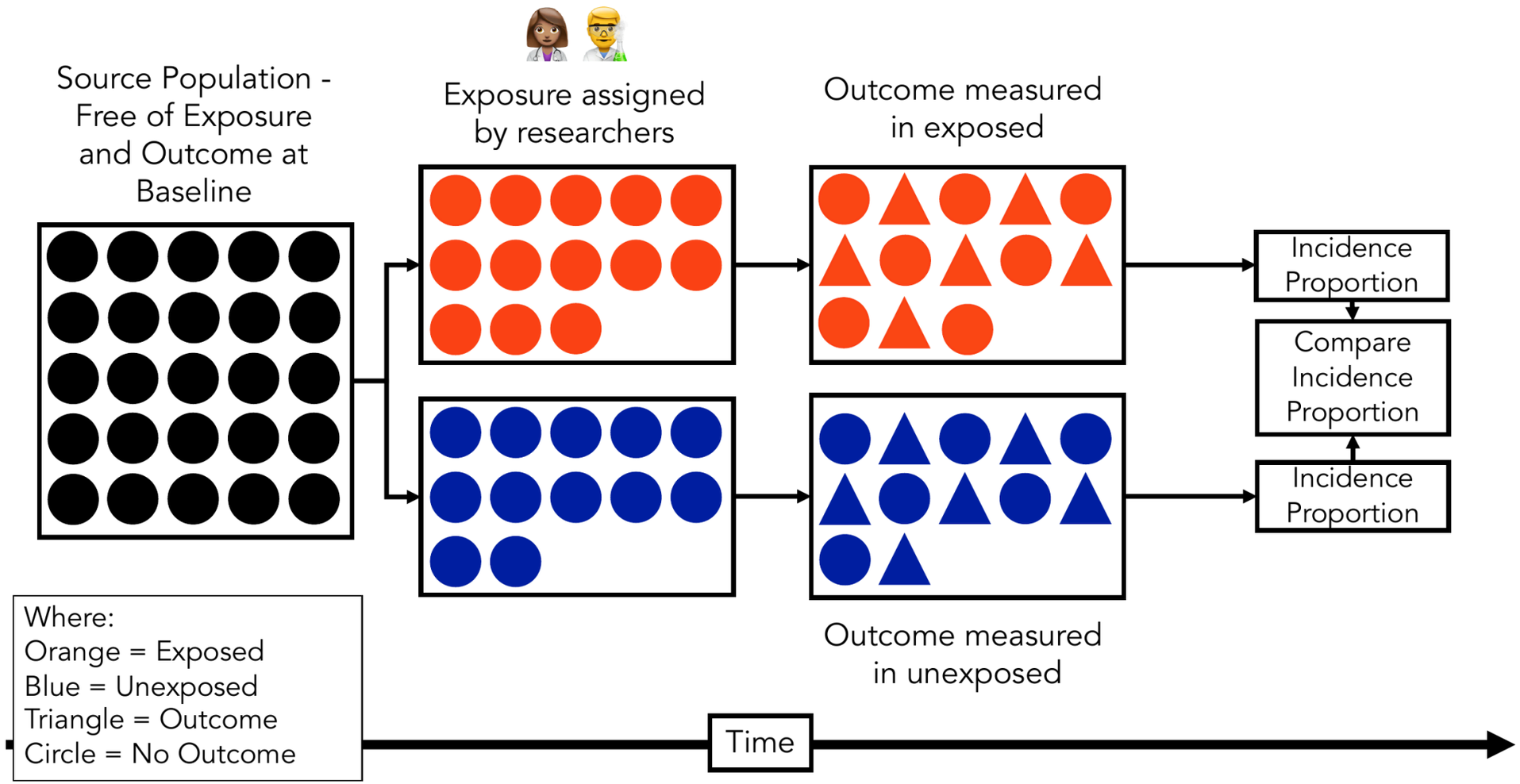
- Often **expensive** and time-consuming
- May have **limited generalizability** to real-world populations
- Not always **ethical or feasible**

What Is a Cohort?

A **cohort** is an identified group of people whose membership is defined by a shared characteristic or event.

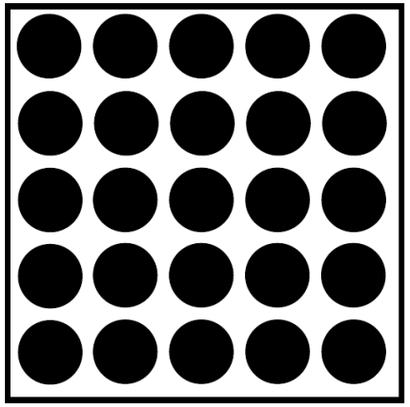
- **Example:** Everyone who graduated from TCU in 2026.
- We can identify **who belongs to the group**.
- In epidemiology, we often follow a cohort over time to observe health outcomes.
- **Key difference between a population and a cohort**: Every member of the cohort is enumerated (i.e., known or listed). That is not necessarily true of a population.

Basic Experimental Study Design



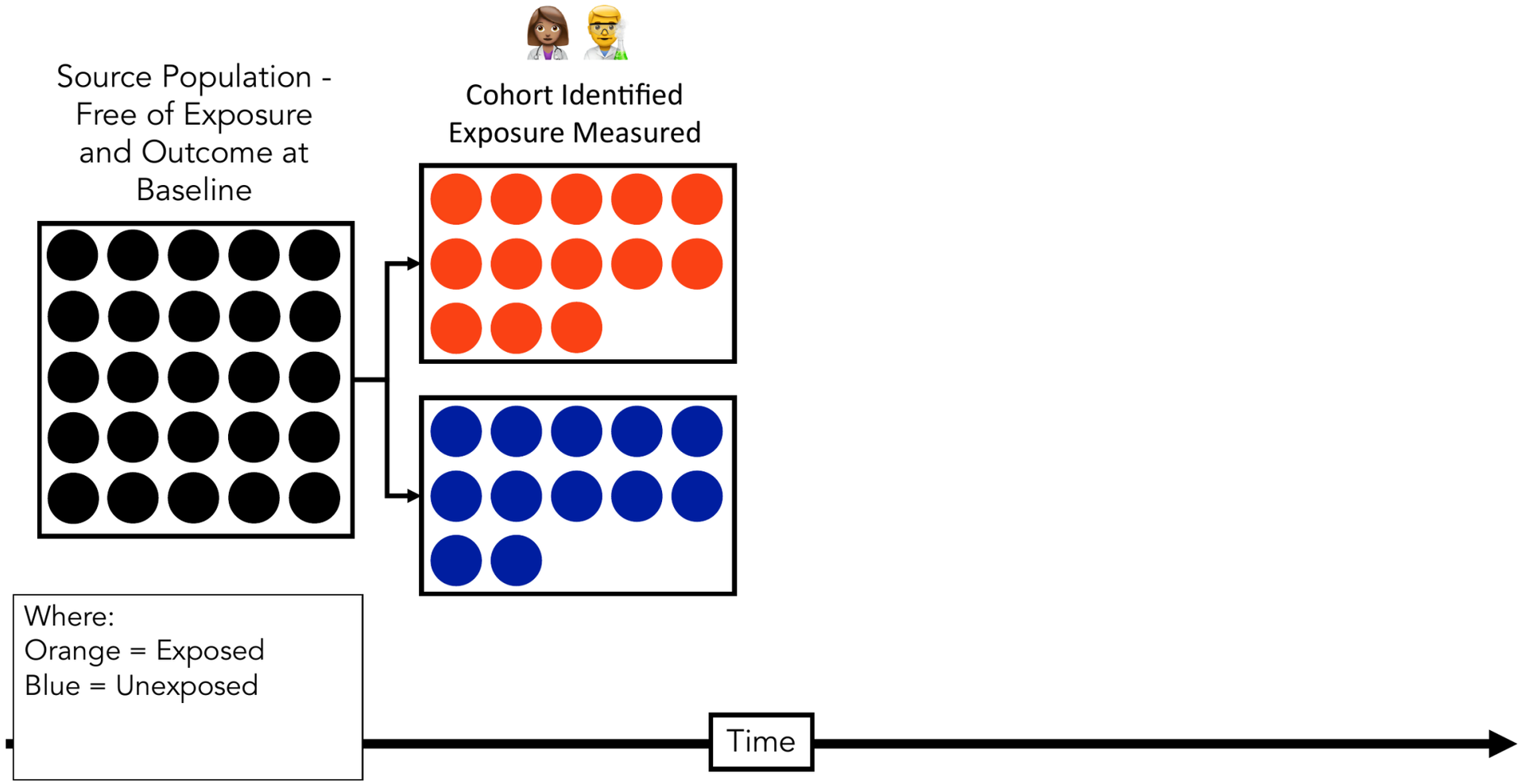
Common Prospective Cohort Study Design

Source Population - Free of Exposure and Outcome at Baseline

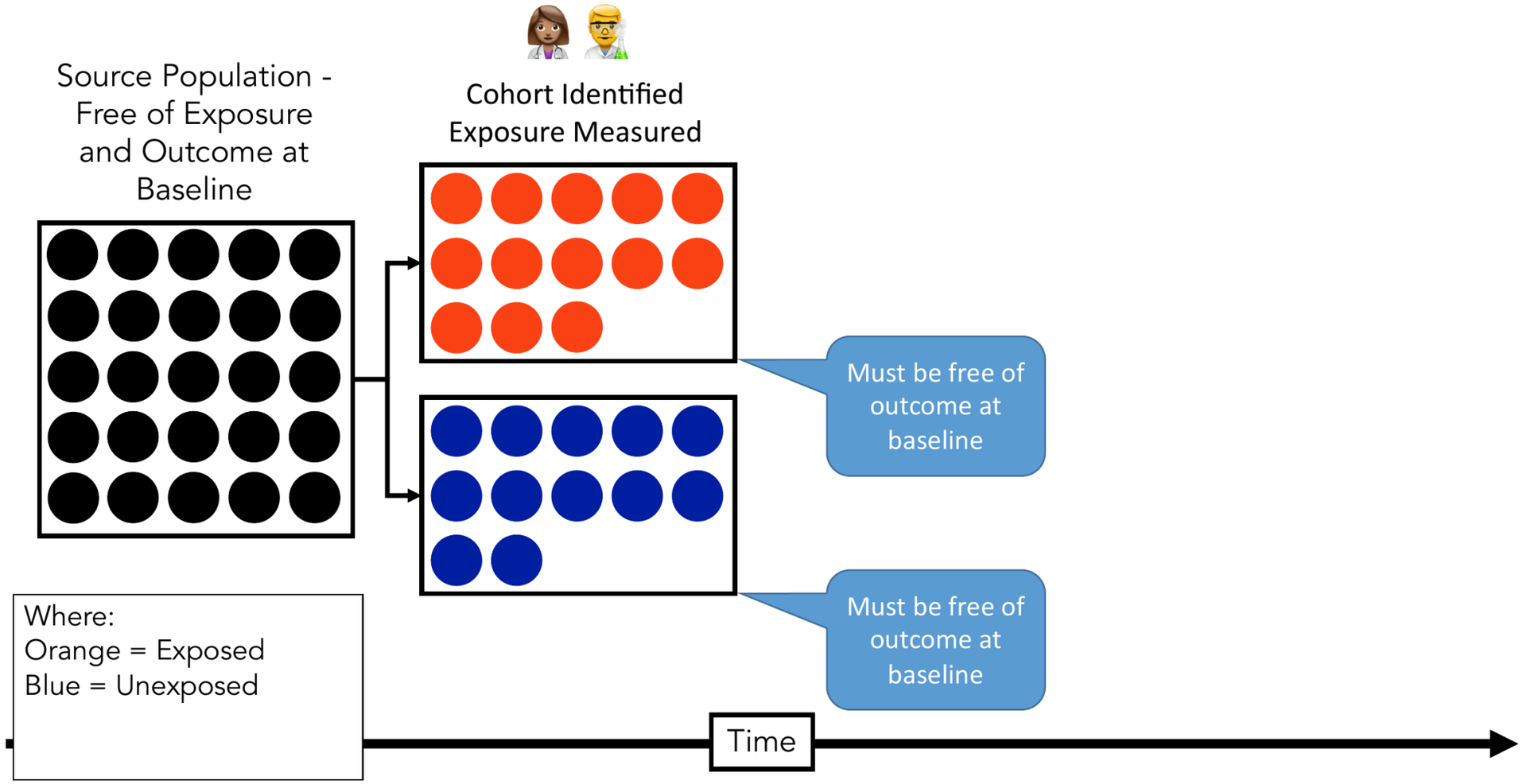




Common Prospective Cohort Study Design



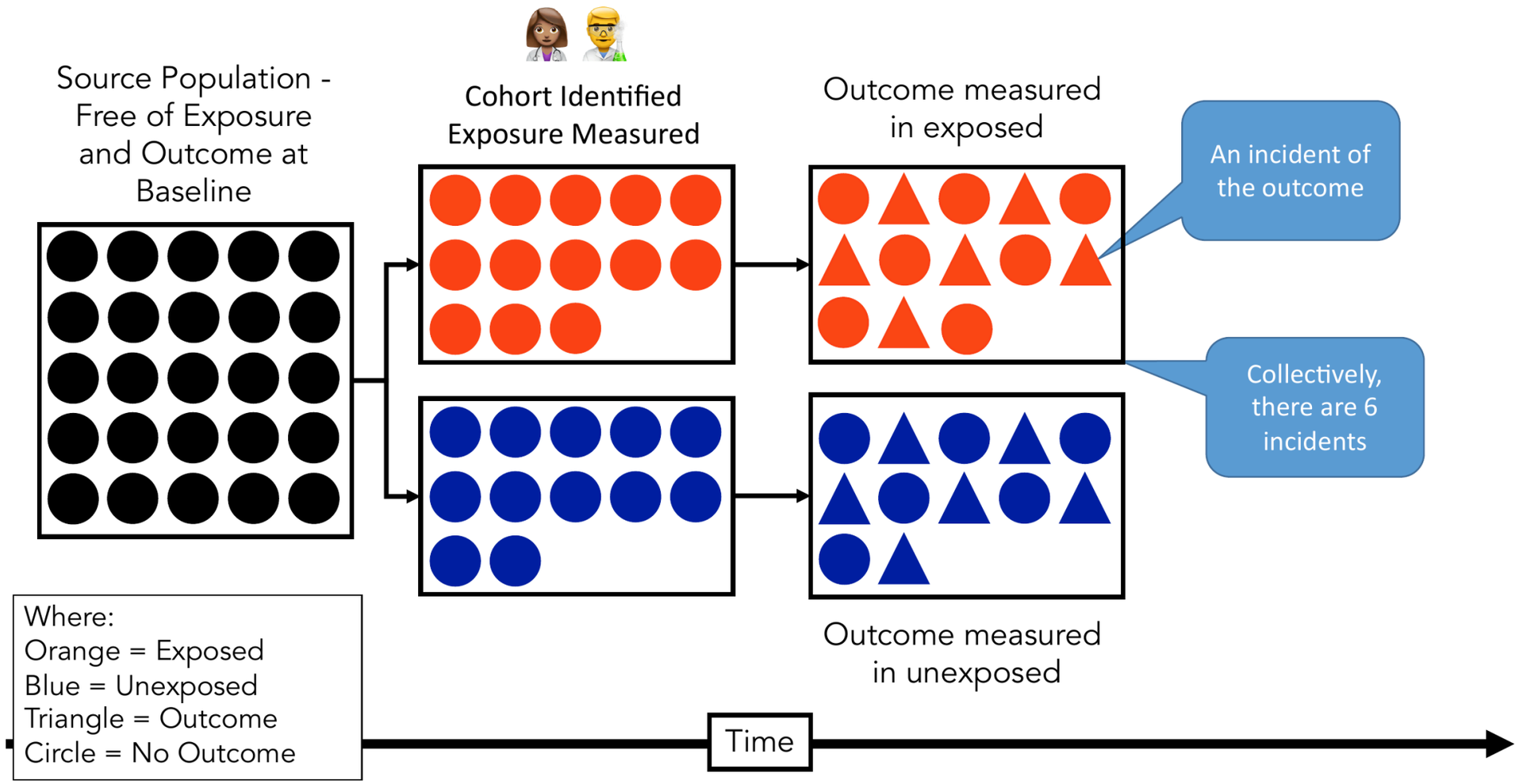
Common Prospective Cohort Study Design



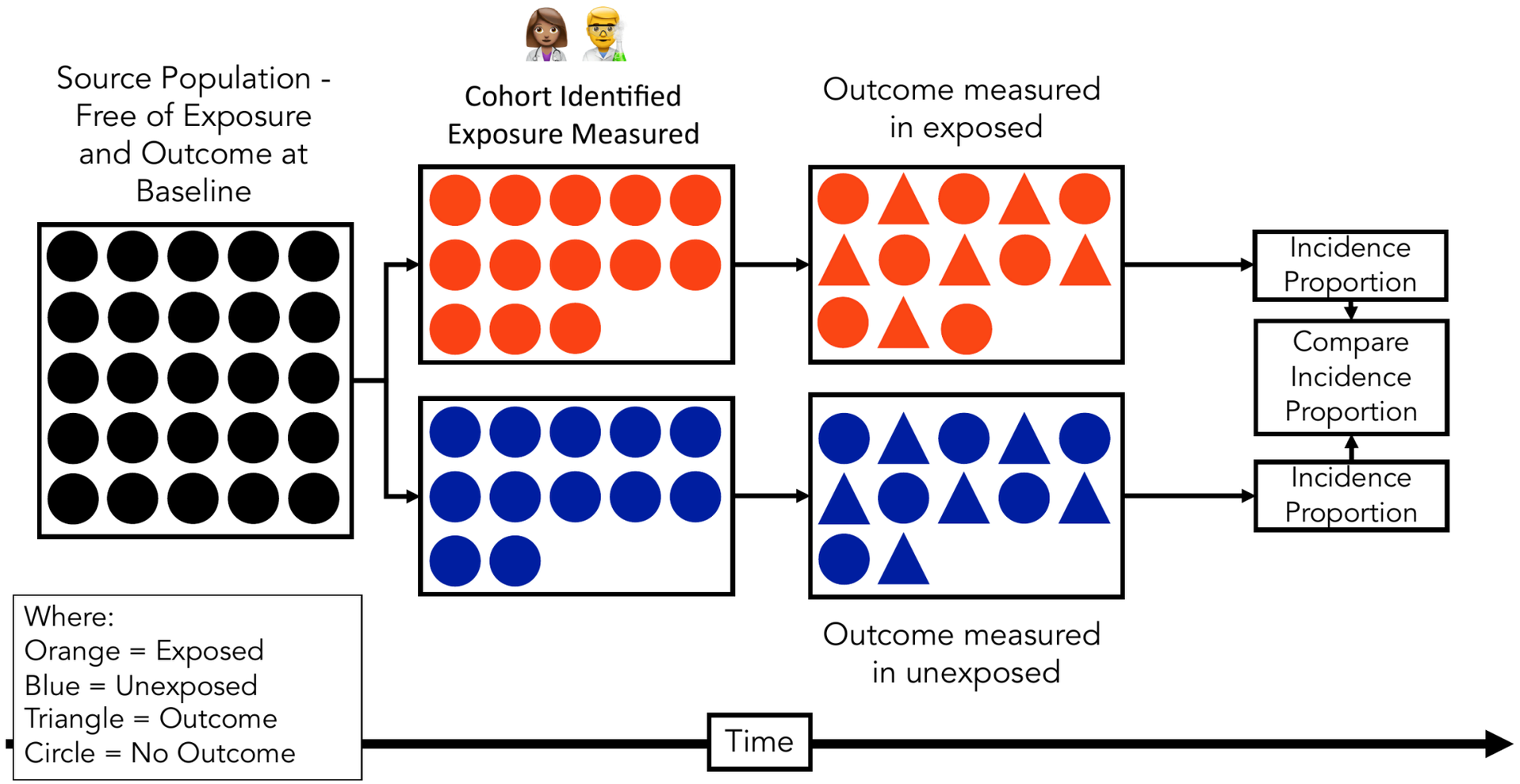
Population at Risk

- All members of the cohort should be **at risk of developing** the outcome of interest.
- Implies that all members should be **free of the outcome** of interest at the outset of follow-up.

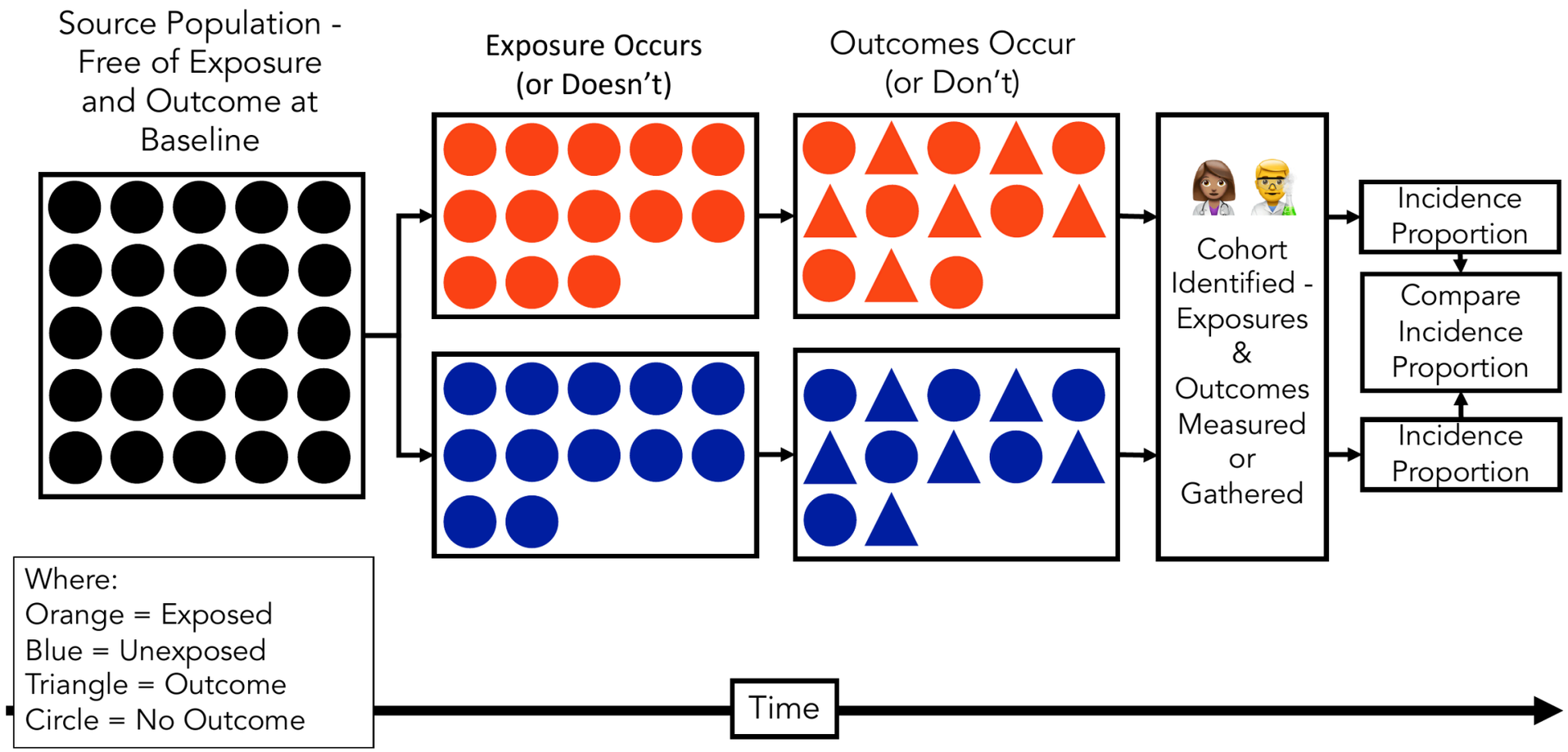
Common Prospective Cohort Study Design



Common Prospective Cohort Study Design



Common Retrospective Cohort Study Design



Prospective vs. Retrospective Cohorts

- Both follow the same direction: **Exposure → Outcome**
- Both sample on the **Exposure**
- **Prospective Cohort**
  - **Exposure measured in the present**
  - Outcomes occur in the future
- **Retrospective Cohort**
  - **Exposure and outcome already occurred**
  - Uses existing records
  - Limited by data quality and missing variables
- **Key Difference:** Timing of data collection — not sampling design.

When to Conduct a Cohort Study

- **Exposure is rare:** efficient for studying uncommon exposures
- **Incidence must be measured:** can directly calculate risk and RR
- **Temporal sequence is important:** exposure clearly precedes outcome
- **Multiple outcomes are of interest:** one exposure → several possible diseases
- **Follow-up is feasible:** resources, time, and stable population available
- **Design Logic:** Cohort studies are strongest when you need clear **exposure → outcome** tracking over time.

The Framingham Heart Study

- **Study Design:** Prospective **Cohort Study**
- **What Was Done:**
  - Enrolled **5,209 adults** to study the development of **cardiovascular disease (CVD)**
  - Measured baseline exposures (e.g., blood pressure, cholesterol, smoking)
  - Followed participants over time for **incident CVD**
- **Why This Design Was Appropriate:**
  - Needed to measure **incidence**
  - Required clear **exposure → outcome** sequence
  - Allowed one exposure to be linked to **multiple outcomes**
  - Long-term follow-up was essential
- **Key Lesson:** Cohort studies are powerful for identifying **risk factors over time**.

Potential Biases in Cohort Studies

- **Selection Bias**
  - **Loss to follow-up (attrition):** selective dropout can cause **attrition bias**
  - *Example:* Participants who developed severe illness were less likely to return for follow-up visits → risk may be underestimated.
- **Information Bias**
  - Misclassification of **exposure**
  - *Example:* Smoking status or diet self-reported inaccurately at baseline.
- **Confounding**
  - A third variable related to both exposure and outcome
  - *Example:* Age is associated with higher blood pressure and higher CVD risk.
- **Key Question:** Which bias is most important in a long prospective cohort?

Assessing Risk in Cohort Studies

- **Risk (Incidence Proportion):** Number of new cases ÷ population at risk
- **Relative Risk (RR):** Risk in exposed ÷ Risk in unexposed
- **Absolute Risk Difference (ARD):** Risk in exposed – Risk in unexposed
- **Key Distinction:**
  - RR shows strength of association
  - ARD shows public health impact

A Prospective Cohort Study Example

- **Research Question:** Is passive tobacco smoke exposure associated with increased respiratory infections in children?
- **Study Design:**
  - Enroll **disease-free children (<12 years)**
  - Classify by **household smoking exposure**
    - Exposed vs Not exposed
- **Follow forward in time**
  - Measure **incident infections**
- **Potential Confounders:**
  - Air pollution
  - Socioeconomic status
  - Crowded housing
- **Next Step:** Calculate **Risk**, **RR**, and **ARD**

Reading a 2×2 Table

	Infection (+)	Infection (–)	Total
Exposed	A	B	A + B
Not Exposed	C	D	C + D

- **Rows** identify exposure status; **columns** identify infection status.
- Each letter represents a **count of people**. For example, **A** is the number of exposed children who developed an infection.
- **A + B** is the total exposed group; **C + D** is the total unexposed group.
- These are the denominators for calculating risk in each group.

Exercise: Risk (Incidence Proportion)

	Infection (+)	Infection (–)	Total
Exposed	100	400	500
Not Exposed	30	470	500

Calculate Risk (Incidence Proportion)

- Risk in the exposed = A / (A + B)
- Risk in the unexposed = C / (C + D)

Interpretation: Risk (Incidence Proportion)

- **Risk in the exposed = 100 / 500 = 0.20 = 20%**
  - 20 out of every 100 exposed children developed an infection during follow-up.
- **Risk in the unexposed = 30 / 500 = 0.06 = 6%**
  - 6 out of every 100 unexposed children developed an infection during follow-up.

Exercise: Relative Risk (RR)

	Infection (+)	Infection (–)	Total
Exposed	100	400	500
Not Exposed	30	470	500

Calculate Relative Risk (RR)

- RR = Risk in the exposed ÷ Risk in the unexposed

Interpretation: Relative Risk (RR)

- **RR = 0.20 ÷ 0.06 ≈ 3.33**
- Exposed children had **about 3.33 times the risk** of infection compared with unexposed children during follow-up.
- Passive smoke exposure was associated with a **substantial increase in risk** during follow-up.

Exercise: Absolute Risk Difference (ARD)

	Infection (+)	Infection (–)	Total
Exposed	100	400	500
Not Exposed	30	470	500

Calculate Absolute Risk Difference (ARD)

- ARD = Risk in the exposed – Risk in the unexposed

Interpretation: Absolute Risk Difference (ARD)

- **ARD = 0.20 – 0.06 = 0.14 = 14 percentage points**
- There were **14 more infections per 100 children** in the exposed group than in the unexposed group during follow-up.
- The risk difference describes the **excess burden of disease** associated with passive smoke exposure.

Poll: Interpreting Relative Risk

In a cohort study examining **smoking and cardiovascular disease**, the Relative Risk (RR) for smokers compared to non-smokers is **1.50**. What does this mean?

A. Smokers have a 50% lower risk of CVD

B. Smokers have a 50% higher risk of CVD

C. Smokers and non-smokers have the same risk

D. Smoking prevents cardiovascular disease

Poll: Interpreting Risk Difference

In a cohort study of **smoking and cardiovascular disease (CVD)**, the risks over the same follow-up period are:

- Risk in smokers = **20%;** Risk in non-smokers = **10%**
- Absolute Risk Difference (ARD) = **20% – 10% = 10 percentage points**

What does this ARD mean?

A. There were 10 more CVD cases per 100 smokers than per 100 non-smokers during follow-up

B. Smoking increases CVD risk by 10 times compared to non-smokers during follow-up

C. There was no association between smoking and CVD during follow-up

D. 10 out of every 100 cases of CVD were caused by smoking during follow-up

Cohort Studies: Strengths and Limitations

Strengths

- Directly measure **incidence**
- Establish clear **temporal sequence** (Exposure → Outcome)
- Study **rare exposures** efficiently
- Examine **multiple outcomes** from one exposure

Limitations

- **Time-consuming** and costly
- Vulnerable to **loss to follow-up**
- Potential **exposure misclassification**
- Less efficient for **rare diseases**